

Daniel Larson

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EDUCATION

University of California, Santa Barbara

Bachelor of Science in Statistics and Data Science

Santa Barbara, CA

Sept. 2024 – Jun. 2027

- Relevant Courses: Principles of Data Science, Probability and Statistics, Regression Analysis, Applied Stochastic Processes, Statistical Learning, Design and Analysis of Experiments.

College of the Canyons

General Education

Santa Clarita, CA

Aug. 2021 – Jun. 2024

EXPERIENCE

Warehouse Specialist

Dorsal

Feb. 2024 – Oct. 2024

Santa Clarita, CA

- Increased production efficiency by examining material preparation, contributing to a smoother assembly flow and reducing downtime.
- Consistently met shipment deadlines by collaborating with team members and adapting quickly to fluctuating order volumes.

PROJECTS

SpiceRack | *Python, Scikit-learn, Flask, TF-IDF, K-Means, SQLite*

Oct. 2025 – May. 2026

- Led a team of developers as project manager to design and deliver a recipe recommendation web app, coordinating task delegation and technical decisions across the full development lifecycle.
- Architected a TF-IDF vectorization pipeline with IDF² boosting and TruncatedSVD compression trained on 1.7M+ recipes, achieving a silhouette score of 0.80 across ~475 clusters to ensure high-quality recommendations.
- Engineered a top-K cluster search with cosine similarity ranking and thread-safe Unsplash API image caching, reducing recommendation latency and enabling seamless AJAX-powered interactions.
- Selected as a finalist in the UCSB Data Science Club Project Showcase out of 100+ competing student teams.

MLB Pitcher Injury Risk Prediction | *R, XGBoost, K-Means, tidymodels* Jan. 2026 - Mar. 2026

- Analyzed 30 years of MLB pitching data from the Lahman Database to predict which pitchers were at risk of injury the following season.
- Built and compared 4 machine learning models (logistic regression, elastic net, random forest, XGBoost), finding that total innings pitched – including postseason – was the leading risk factor.
- Applied K-means clustering to group pitchers into 3 archetypes by workload and mechanics; the highest-workload group had nearly double the injury rate (26.6%) compared to the lowest (14.1%).
- Optimized the model to flag 80% of at-risk pitchers correctly; the top 10% of highest-risk pitchers were injured at a 53.6% rate, nearly 3 times the 19% average.

CLUBS

Data Science Club (UCSB) | *Python, Pandas, NumPy, SQL*

Oct. 2024 – Present

- Won “Best Sports Analyst Project” at Datathon 2025 and was selected as a 2025 Project Pitchfire Finalist.
- Strengthened communication and presentation skills through networking events and resume workshops.

Gaucha Gaming (UCSB) | *Community Engagement Intern*

Jan. 2026 – Present

- Collaborated with officers to support community initiatives, online events, and engagement efforts to improve member experience and retention.
- Represented the organization professionally while helping maintain a safe and inclusive community space.

SKILLS

Languages: Python, R, SQLite, C++

Libraries and Frameworks: Pandas, NumPy, Matplotlib, Scikit-Learn, XGBoost, tidymodels

Tools: Git, GitHub, VS Code, PyCharm, Jupyter Notebook

Achievements: Datathon 2025 – Best Sports Analyst Project, Anti-DUI – Project Showcase Finalist